S^{2}CLNet: Structure-Constrained Semantic Contrastive Learning for Referring Remote Sensing Image Segmentation

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***Abstract*—Referring remote sensing image segmentation (RRSIS) aims to segment target ground objects in high-resolution remote sensing imagery according to textual descriptions. Due to complex backgrounds and large-scale variations, accurately aligning linguistic semantics with spatial regions remains a challenging problem. Existing RRSIS methods are predominantly segmentation-centric, where textual descriptions mainly serve as conditional guidance for pixel-wise prediction rather than explicitly enforcing semantic consistency between visual regions and linguistic representations. This limitation often leads to suboptimal fine-grained alignment and insufficient suppression of background-dominant regions. To address these challenges, we propose a novel RRSIS framework centered on structure-constrained semantic contrastive learning (CL), termed S^{2}CLNet. Departing from the segmentation-driven paradigm, instead of relying solely on segmentation supervision, S^{2}CL introduces region-level structural constraints to perform fine-grained CL in the joint embedding space. This design encourages semantically corresponding visual regions and linguistic representations to be more closely aligned while effectively separating target regions from irrelevant background areas. Furthermore, we develop a dynamic modality balancing module (DMBM) to enhance cross-modal interaction under complex remote sensing scenarios. The DMBM jointly models bidirectional cross-modal attention (BCA) and dynamically adjusts the relative contributions of visual and linguistic modalities according to scene complexity and linguistic specificity, thereby facilitating more adaptive and robust multimodal feature integration. Extensive experiments on three public benchmarks, RefSegRS, RRSIS-D, and RISBench, demonstrate that the proposed method achieves superior performance compared with state-of-the-art approaches. The code will be publicly available at https://github.com/45degreesl**

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I. Introduction

REFERRING remote sensing image segmentation (RRSIS) aims to locate and segment the corresponding ground target in remote sensing imagery according to a given textual description [1], [2], [3], [4], [5]. By allowing users to specify regions of interest through natural language, this technique significantly facilitates the extraction of specific objects and enhances the efficiency of remote sensing data interpretation [6], [7], [8], [9], [10]. With the rapid development of multimodal learning, RRSIS plays a critical role in a variety of remote sensing tasks, such as land-use analysis, environmental monitoring, and disaster management [11], [12], [13], [14], [15], [16].

Over the past two years, RRSIS has received increasing attention, benefiting from rapid progress in vision–language pretraining, multimodal representation learning, and the growing demand for fine-grained remote sensing image interpretation [17], [18], [19], [20]. Most existing methods are built upon an image encoder and a text encoder and perform pixel-wise segmentation of specific targets through different image–text alignment paradigms. Specifically, RMSIN [2] leverages interscale and cross-scale interactions to achieve feature fusion across the vision–language modality. FIANet [3] employs a fine-grained image–text alignment to perceive the ground objects and spatial position. Li et al. [21] designed a bidirectional alignment module to refine multimodal features via vision–language and language–vision interactions in the encoder. These approaches have advanced the field by enabling flexible and referring segmentation under natural language guidance.

However, current RRSIS approaches are dominated by a segmentation-centric learning paradigm, where multimodal fusion is primarily optimized through segmentation losses to predict the mask toward the ground-truth annotation. In this setting, textual descriptions are mainly used as conditional cues to guide pixel-wise prediction, rather than being explicitly constrained to form discriminative semantic correspondences with visual regions. As a result, the learned image–text representations tend to emphasize local correspondence for mask

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Fig. 1. Visual comparisons of the conventional CL and the proposed S^{2}CL on the RefSegRS test set. Both of these are integrated into the same baseline architecture.

prediction, while lacking a global, structured constraint that enforces cross-modal semantic consistency.

Contrastive learning (CL) provides an effective mechanism for enforcing global semantic alignment between visual and textual representations by explicitly pulling matched image–text pairs closer while pushing mismatched ones apart in embedding spaces. CRIS [22] introduces contrastive objectives into referring image segmentation and demonstrates that image–text CL can enhance semantic discriminability in natural images. However, directly transferring image-level CL paradigms to RRSIS remains challenging. Remote sensing imagery is characterized by large-scale variations, arbitrary object orientations, and background-dominant scenes, where target regions often occupy only a small fraction of the image. Consequently, conventional image-level contrastive objectives tend to introduce semantic ambiguity, suffer from unreliable negative sampling, and fail to establish precise region-aware semantic alignment, thereby limiting its effectiveness for RRSIS.

To overcome these limitations, we propose a structure-constrained semantic CL (S^{2}CL) framework, termed S^{2}CLNet, which enforces fine-grained vision–language alignment under explicit structural constraints by leveraging region masks as structure priors to construct semantically consistent positive and discriminative negative pairs. As illustrated in Fig. 1, compared with conventional CL using the same baseline architecture, the proposed S^{2}CL achieves more accurate localization and better boundary preservation. Furthermore, to dynamically balance the uneven contributions of visual and textual modalities across different scenes, we introduce a dynamic modality balancing module (DMBM) that performs bidirectional cross-modal attention (BCA) with adaptive modality weighting, enabling more context-aware multimodal fusion in complex remote sensing scenarios.

The contributions of this work are summarized as follows.

1) We propose an S^{2}CL framework for RRSIS. Unlike existing segmentation-centric approaches, S^{2}CL introduces structure-constrained spatial supervision to perform fine-grained CL, enabling precise semantic alignment between textual descriptions and target regions while suppressing background interference.
2) We design a DMBM to enhance vision–language interaction in complex remote sensing scenes. This module explicitly models BCA and dynamically adjusts the relative contributions of visual and textual cues, thereby facilitating more balanced and context-aware multimodal feature fusion under complex and diverse remote sensing scenes.
3) Extensive experiments on three public RRSIS benchmarks demonstrate the effectiveness and superiority of the proposed S^{2}CLNet, and comprehensive ablation studies further verify the effectiveness of these designs.

The remainder of this article is organized as follows. Section II provides a concise overview of the background and related work. In Section III, we present a detailed description of the proposed method and its key components. Section IV reports extensive experimental results on three public remote sensing datasets, including comparative evaluations and ablation studies. Finally, we conclude the proposed method in Section V.

II. Background and Related Work

A. Referring Image Segmentation for Natural Images

Referring image segmentation aims to generate a pixel-wise mask for a target object specified by a natural language expression. Early studies laid the foundation of this task by jointly modeling visual and linguistic information. Hu et al. [23] first proposed a CNN-LSTM framework, establishing a standard pipeline consisting of visual feature extraction, multimodal interaction, and mask prediction. Building upon this paradigm, Liu et al. [24] introduced a gated recurrent mechanism to better capture the temporal dependencies in language representations, leading to improved cross-modal alignment. Subsequent works focused on enhancing multimodal feature fusion and instance-level reasoning. Margffoy-Tuay et al. [25] proposed a dynamic multimodal instance segmentation framework, where language queries are used to guide feature fusion, enabling more adaptive vision–language matching. Li et al. [26] further introduced a recurrent refinement network to iteratively integrate multiscale visual features, improving segmentation accuracy for objects with varying sizes. To better exploit hierarchical visual cues, later approaches incorporated multiscale and gated fusion strategies. Many methods [27], [28], [29] enhance the interaction between low-level spatial details and high-level semantic representations through hierarchical modeling and gating mechanisms, leading to more robust segmentation in complex scenes.

Meanwhile, attention-based cross-modal interaction has become a dominant paradigm. Ye et al. [30] pioneered the

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use of cross-modal self-attention to capture long-range dependencies between visual and linguistic features. Luo et al. [31] proposed cascaded group attention to model instance-level semantic differences via stepwise reasoning. Hu et al. [32] further introduced bidirectional relationship reasoning to explicitly model mutual guidance between vision and language. Some works [22], [33] have leveraged refined attention mechanisms and stronger vision–language pretraining to enhance cross-modal alignment and improve referring segmentation performance. DETRIS [34] addresses cross-modal misalignment between independently pretrained encoders through a dense aligner and a text adapter. More recently, Gholami et al. [35] proposed the dynamic multimodal instance segmentation framework that exploits language queries to facilitate adaptive instance-aware representation learning. ReferSAM [36] enhances SAM for referring image segmentation via fine-grained vision–language alignment and improved prompt encoding.

B. Remote Sensing Referring Image Segmentation

RRSIS is a vision–language segmentation task that identifies target objects in remote sensing images according to referring expressions. Recent advances in vision–language pretraining and multimodal representation learning have greatly promoted the development of this task [37], [38], [39], [40], [41], [42]. Yuan et al. [1] formally defined the RRSIS task and constructed the first benchmark dataset, RefSegRS, establishing a foundational research setting for this field. They further proposed a language-guided cross-scale enhancement strategy to facilitate multimodal interaction. Subsequently, Liu et al. [2] introduced the larger scale RRSIS-D dataset and designed a multiscale interactive network to accommodate the significant variations in target scale and orientation commonly observed in remote sensing imagery. FIANet [3] is a fine-grained image–text alignment framework that decouples textual semantics from spatial relationships, thereby enhancing the discriminative capability of cross-modal features.

More recent studies have further advanced RRSIS by integrating foundation models and improving generalization in complex scenes. Chen et al. [11] pioneered the joint utilization of CLIP and SAM, leveraging their cross-modal alignment and universal segmentation capabilities to establish a new application paradigm for RRSIS. Zhang et al. [43] introduced bidirectional spatial correlation modules to strengthen cross-modal interaction and enhance target–background discrimination. Chen et al. [4] proposed a decoupled "coarse localization-fine segmentation" framework to improve robustness under complex scene conditions. Furthermore, CroBIM-U [44] introduces a referring uncertainty map to characterize spatial uncertainty, which is subsequently utilized to guide adaptive inference in a spatially aware manner.

C. CL in Remote Sensing

CL has emerged as an effective representation learning strategy for remote sensing image analysis and has demonstrated remarkable potential across a wide range of remote sensing applications. Biswas and Tešić [45] introduced a support-guided debiased CL framework for remote sensing object detection, which mitigates pseudo-label noise and false-negative bias through support-set-assisted pseudo-label generation and a debiased contrastive objective. Liu et al. [46] developed an unsupervised contrastive hashing method that combines cross autoencoders with a multiobjective loss to improve semantic consistency for cross-modal retrieval. Lv et al. [47] constructed a graph-based CL network to address modal heterogeneity in heterogeneous remote sensing change detection by leveraging superpixel graph structures and contrastive objectives. To enhance self-supervised remote sensing representation learning, Stival et al. [48] proposed semantically aware CL (SACo+), which integrates semantic band combinations and texture-based representations into the CL process. Moreover, He et al. [49] introduced a semi-supervised balanced CL framework for hyperspectral and LiDAR classification, where balanced pseudo-label generation and CL are jointly exploited to enhance multimodal feature representations. Li et al. [50] designed a feature comparison detector for small and weak target detection, enhancing target–background discrimination through spatially guided feature alignment and instance-controlled label assignment.

Despite these advances, existing CL approaches in remote sensing are largely designed for unimodal or weakly multimodal tasks, and their extension to fine-grained vision–language problems remains limited. In particular, how to effectively integrate CL into RRSIS, where precise spatial location and semantic alignment between textual descriptions and visual regions are simultaneously required, has not yet been investigated.

III. Methodology

A. Overview of the Proposed Method

In this article, we propose a novel RRSIS method S²CLNet, and the overall framework is demonstrated in Fig. 2. Following earlier works [2], [43], given an input remote sensing image and its paired referring expression, visual and linguistic representations are first extracted using an image encoder and a text encoder, respectively. The image encoder produces hierarchical visual features at multiple stages, capturing rich spatial structures and multiscale semantics that are essential for segmenting objects with large appearance variations in remote sensing scenes. In parallel, the text encoder encodes the referring expression into a global linguistic embedding that conveys semantic category and spatial information.

To facilitate robust interaction between visual and linguistic features, we design a DMBM, which enables bidirectional information exchange and adaptive balancing of visual and textual cues. This design allows the model to capture complementary multimodal representations that jointly encode spatial details and language-guided object semantics. The fused features are then progressively integrated by a segmentation decoder to produce the final pixel-wise prediction. In addition, we propose the S²CL to strengthen cross-modal semantic consistency during training. By exploiting ground-truth segmentation masks to guide patch-level contrastive supervision, S²CL explicitly enforces semantic alignment between

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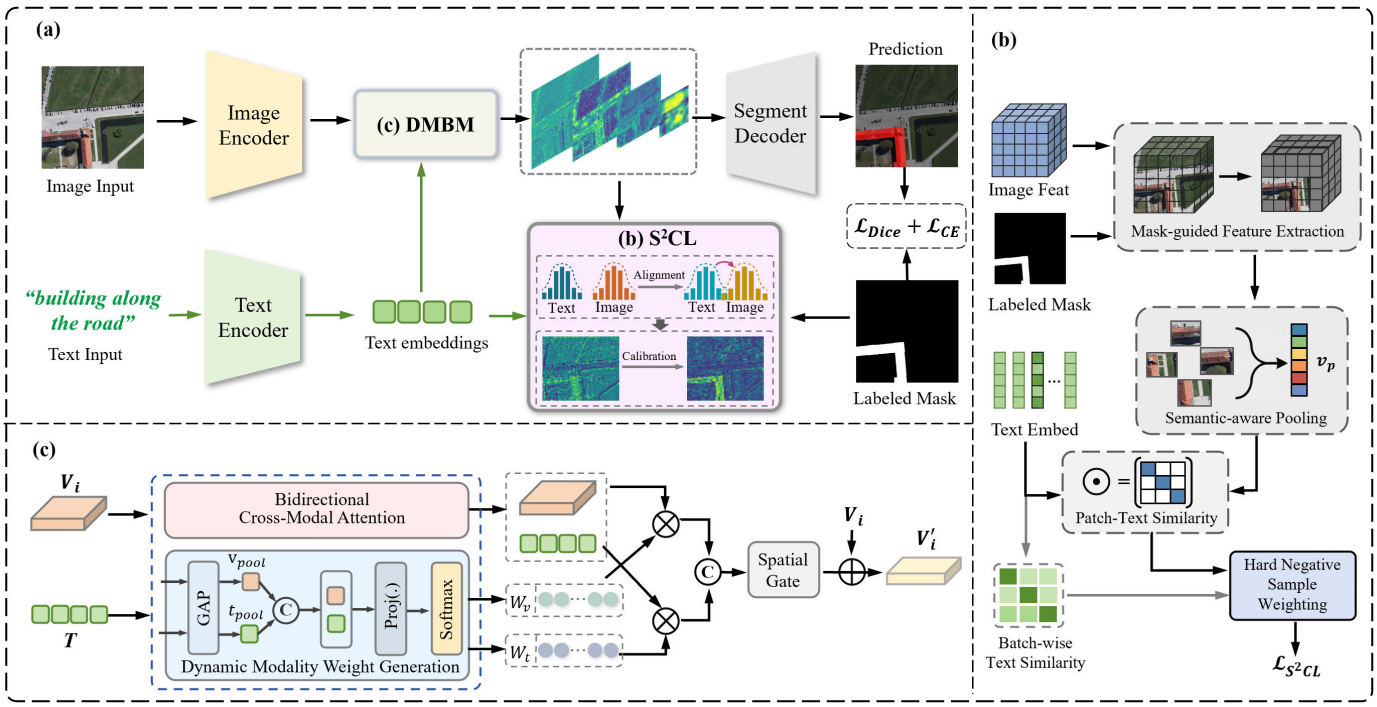


Fig. 2. Illustration of (a) framework of the proposed method, (b) structure-constrained semantic CL (S^{2}CLNet), and (c) DMBM.

textual descriptions and target regions while suppressing background interference. This contrastive formulation enhances the discriminative power and robustness of the learned representations, leading to improved segmentation performance in complex remote sensing scenes. Detailed descriptions of S^{2}CL and DMBM are provided in Sections III-B and III-C.

B. Structure-Constrained Semantic CL

Unlike conventional CL, the proposed S^{2}CL introduces structure-constrained patch-level supervision to explicitly align textual semantics with foreground regions. By incorporating segmentation priors and semantic-aware hard negative weighting, S^{2}CL enhances cross-modal discriminability and robustness in large-scale, complex remote sensing scenes.

Given a batch of images and paired textual descriptions, the extracted multimodal visual features and linguistic embeddings are denoted as $\boldsymbol{V} \in \mathbb{R}^{B\times H\times W\times C_v}$ and $\boldsymbol{T} \in \mathbb{R}^{B\times L\times C_t}$, where B denotes the batch size, and H, W, and C_v are the height, width, and channel number of the image features. L is the number of text tokens, and C_t represents the text embedding dimension. The overview of the S^{2}CL is illustrated in Fig. 2(b), with detailed descriptions provided in the subsequent section. Algorithm 1 provides the pseudocode of S^{2}CL, where the main components are involved.

1) GP and VFS: We design a grid partition (GP) to localize CL at a suitable spatial granularity for remote sensing imagery. Compared with image-level or pixel-level formulations, GP provides a stable and controllable unit for contrastive supervision, which is particularly important when the target regions occupy only a small portion of large-scale scenes. By further selecting foreground patches of visual features, S^{2}CL effectively suppresses background interference and ensures that CL focuses on semantically meaningful regions.

For each image batch, the corresponding foreground mask is denoted as $\mathbf{M}$. To ensure spatial alignment with the visual feature map, the mask is first interpolated as

$$\hat{\mathbf{M}} = \text{Interp}(\mathbf{M}), \quad \hat{\mathbf{M}} \in \mathbb{R}^{B\times 1\times H\times W}. \tag{1}$$

The visual feature map $\boldsymbol{V}$ is then partitioned into a set of nonoverlapping feature patches $\boldsymbol{V}_k \in \mathbb{R}^{B\times(H/g)\times(W/g)\times C_v}, k = 1,\ldots,g^2$ according to a predefined grid number g. For the ith sample and the kth patch, the foreground ratio is computed as

$$\rho_k^i = \frac{\sum_{(u,v)\in \boldsymbol{V}_k^i} \hat{\mathbf{M}}^i(u,v)}{|\boldsymbol{V}_k^i|} \tag{2}$$

where (u, v) denotes the spatial position within a patch and $|\boldsymbol{V}_k^i|$ denotes the total number of spatial pixels within the kth patch of the ith sample. Only patches satisfying $\rho_k^i > \eta$ are retained as valid foreground regions. Let $\mathcal{R}$ denote the set of valid region indices, where the threshold η is empirically set to 0.1.

2) Semantic-Aware Average Pooling: For each selected foreground-dominant region, we further design a semantic-aware average pooling to aggregate visual features. Specifically, this operation emphasizes semantically consistent foreground regions during feature aggregation by restricting the pooling process to mask-consistent areas. As a result, it produces more robust region-level representations while effectively suppressing background interference. For each valid region $r \in \mathcal{R}$ and sample i, these background responses are suppressed through mask-guided feature refinement

$$\tilde{\boldsymbol{V}}_r^i = \boldsymbol{V}_r^i \odot \hat{\mathbf{M}}_r^i \tag{3}$$

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Algorithm 1 Pseudocode of S²CL

Input: Visual features $\boldsymbol{V}$, text embeddings $\boldsymbol{T}$, mask $\boldsymbol{M}$
Output: Contrastive loss $\mathcal{L}_{\text{S}^2\text{CL}}$
$\hat{\boldsymbol{M}} \leftarrow \text{Interp}(\boldsymbol{M})$
Partition $\boldsymbol{V}$ into regions $\{\boldsymbol{V}_k\}$
// Visual feature selection
for each $\boldsymbol{V}_k$ **do**
  Compute foreground ratio r_k
  if $r_k > \eta$ **then**
    Add $\boldsymbol{V}_k$ to $\mathcal{R}$
  end if
end for
// Semantic-aware average pooling
for $\boldsymbol{V}_k \in \mathcal{R}$ **do**
  $\tilde{\boldsymbol{V}}_k \leftarrow \boldsymbol{V}_k \odot \hat{\boldsymbol{M}}_k$
  $\boldsymbol{f}_k \leftarrow \text{AvgPool}(\tilde{\boldsymbol{V}}_k)$
end for
$\hat{\boldsymbol{T}} \leftarrow \text{AvgPool}(\boldsymbol{W}_t \boldsymbol{T})$
// Similarity computation
Compute region-text similarity $s_r(i, j)$
Compute batch-wise text similarity $s_t(i, j)$
// Hard negative weighting
for each pair $(i, j), i \neq j$ **do**
  Compute weight $w_{i,j}$ based on $s_t(i, j)$
end for
// Contrastive loss
Compute $\mathcal{L}_{\text{S}^2\text{CL}}$ using $s_r(i, j)$ and $w_{i,j}$
return: $\mathcal{L}_{\text{S}^2\text{CL}}$

where $\odot$ denotes element-wise multiplication and $\hat{\mathbf{M}}_r^i$ is the corresponding mask label to the region of $\boldsymbol{V}_r^i$.

The foreground feature is then obtained by aggregating masked features over the spatial dimension

$$\boldsymbol{f}_r^i = \frac{1}{N_r^i} \sum_{(u,v)\in \boldsymbol{V}_r^i} \tilde{\boldsymbol{V}}_r^i(u, v) \tag{4}$$

where N_r^i denotes the number of foreground pixels within region $\boldsymbol{V}_r^i$.

3) Similarity Computation and HNSW: Here, we will compute two complementary similarity measures, namely patch-text similarity and batch-wise text similarity. The former quantifies the alignment between patch-level visual features and textual representations, whereas the latter captures semantic relationships among textual descriptions within a mini-batch. Both similarities are leveraged to formulate the CL loss.

To ensure dimensional consistency with visual features, the textual embeddings are projected to the same feature dimension via a learnable linear transformation

$$\hat{\boldsymbol{T}} = \text{Avg_Pool}(\boldsymbol{W}_t \boldsymbol{T}), \quad \hat{\boldsymbol{T}} \in \mathbb{R}^{B\times C_v} \tag{5}$$

where $\boldsymbol{W}_t$ is a learnable projection matrix and Avg_Pool denotes the operation of average pooling.

Then, the patch-text similarity $s_r(i, j)$ and batch-wise text similarity $s_t(i, j)$ can be obtained as follows. To avoid ambiguity, we use k to index an original grid patch, r to index a valid foreground-dominant region, and i and j to index samples in the mini-batch. For each region $r \in \mathcal{R}$, $\mathcal{B}_r$ denotes the set of samples whose foreground ratio in region r is larger than η. The pooled region feature of the ith sample in region r is denoted as $\boldsymbol{f}_r^i$, and the projected text feature of the jth sample is denoted as $\hat{\boldsymbol{t}}^j$. The region-text similarity is defined as

$$s_r(i, j) = \frac{\left(\boldsymbol{f}_r^i\right)^\top \hat{\boldsymbol{t}}^j}{\left\|\boldsymbol{f}_r^i\right\|_2 \left\|\hat{\boldsymbol{t}}^j\right\|_2}, \quad i, j \in \mathcal{B}_r. \tag{6}$$

The batch-wise text similarity is computed by

$$s_t(i, j) = \frac{\left(\hat{\boldsymbol{t}}^i\right)^\top \hat{\boldsymbol{t}}^j}{\left\|\hat{\boldsymbol{t}}^i\right\|_2 \left\|\hat{\boldsymbol{t}}^j\right\|_2}. \tag{7}$$

In the S²CL, a challenge arises from the presence of semantically similar yet visually ambiguous samples, which are particularly prevalent in remote sensing scenarios due to repetitive structures, large-scale backgrounds, and subtle interclass variations. Uniformly weighting all negative text samples ignores their varying semantic relevance. Since easy negatives provide limited supervisory signals while semantically similar negatives are more likely to confuse cross-modal representation learning, treating them equally leads to suboptimal optimization. To address this issue, we introduce a hard negative sample weighting (HNSW) strategy that explicitly accounts for intertext semantic similarity within a batch. By assigning higher weights to semantically closer negative samples, the model is encouraged to focus on resolving fine-grained cross-modal ambiguities, thereby enhancing the robustness and discriminability of the learned patch-text representations.

To explicitly model hard negatives, the batch-wise text similarity s_t is utilized. If the semantic similarity between two text descriptions exceeds a predefined threshold δ, the corresponding mismatched patch-text pair is regarded as a hard negative sample. Its contribution is amplified using an adaptive weighting strategy

$$w_{i,j} = \begin{cases} 1, & \text{if } s_t(i, j) \leq \delta \\ 1 + \dfrac{s_t(i, j) - \delta}{1 - \delta}, & \text{if } s_t(i, j) > \delta. \end{cases} \tag{8}$$

For each anchor sample i, the S²CL loss is obtained by averaging over all valid regions

$$\mathcal{L}_{\text{S}^2\text{CL}} = -\frac{1}{|\mathcal{R}|} \sum_{r\in\mathcal{R}} \log \frac{\exp\left(s_r(i, i)/\tau\right)}{\sum_{j\neq i} w_{i,j} \exp\left(s_r(i, j)/\tau\right)} \tag{9}$$

where τ is a temperature parameter, $\mathcal{R}$ denotes the set of valid foreground-dominant regions, i denotes the anchor sample, j denotes a mismatched text sample, and $w_{i,j}$ adaptively adjusts the weight for different sample pairs. Specifically, given an anchor patch feature $\boldsymbol{f}_r^i$, its corresponding text embedding $\hat{\boldsymbol{t}}^i$ constitutes a positive pair, while the text embeddings $\hat{\boldsymbol{t}}^j (j \neq i)$ from other image–text pairs are regarded as negative samples.

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C. Dynamic Modality Balancing Module

In the RRSIS task, visual and textual modalities often provide complementary yet uneven information. Visual features capture spatial structures and fine-grained details of ground objects, while textual features offer high-level semantic guidance for target localization. However, the naive unidirectional attention mechanism is insufficient to fully exploit cross-modal interactions, and the relative importance of each modality may vary across different scenes and referring expressions. To address this challenge, we propose the DMBM to perform adaptive cross-modal integration by jointly modeling bidirectional vision–language interactions and dynamically modulating the contributions of visual and textual features. This design enables more balanced and context-aware multimodal representations, thereby improving robustness in complex remote sensing scenarios. The detailed structure of DMBM is shown in Fig. 2(c).

Formally, $\boldsymbol{X}_i \in \mathbb{R}^{B\times N_v\times C_v}$ denotes the visual feature embeddings extracted from the image backbone at the ith stage, while $\boldsymbol{L} \in \mathbb{R}^{B\times N_l\times C_l}$ denotes the textual feature embeddings obtained by a pretrained text encoder. Here, N_v and N_l represent the numbers of visual tokens and language tokens, respectively, and C_v and C_l denote their corresponding feature dimensions.

1) Bidirectional Cross-Modal Attention: To facilitate fine-grained interaction and semantic alignment between visual and linguistic modalities, we adopt a BCA, consisting of an image-to-language attention branch and a language-to-image attention branch. By enabling mutual information exchange, this formulation allows each modality to selectively attend to semantically relevant components in the other, rather than relying on a single directional fusion.

In the image-to-language attention branch, visual features are used to guide the aggregation of linguistic information. Specifically, visual tokens act as queries to attend over language tokens, encouraging the textual representation to emphasize words and phrases that are most relevant to the visual content. The resulting language-aware representation is computed as

$$\boldsymbol{A}^i_{v\to l} = \text{Softmax}\left(\frac{\boldsymbol{X}_i\boldsymbol{W}^{vl}_Q(\boldsymbol{L}\boldsymbol{W}^{vl}_K)^\top}{\sqrt{d_k}}\right)\boldsymbol{L}\boldsymbol{W}^{vl}_V \tag{10}$$

where $\boldsymbol{W}^{vl}_Q$, $\boldsymbol{W}^{vl}_K$, and $\boldsymbol{W}^{vl}_V$ are the learnable projection matrices for query, key, and value, respectively, and d_k denotes the dimensionality of the query and key.

The language-to-image attention branch leverages textual features to guide visual feature refinement. Here, language tokens serve as queries to attend over visual tokens, enabling the model to highlight spatial regions that are semantically consistent with the referring expressions. The text-informed visual features are obtained as

$$\boldsymbol{A}^i_{l\to v} = \text{Softmax}\left(\frac{\boldsymbol{L}\boldsymbol{W}^{lv}_Q\left(\boldsymbol{X}_i\boldsymbol{W}^{lv}_K\right)^\top}{\sqrt{d_k}}\right)\boldsymbol{X}_i\boldsymbol{W}^{lv}_V. \tag{11}$$

2) Dynamic Modality Weight Generation: In remote sensing images, the contribution of visual and textual features may vary significantly across samples due to varying background complexity, object size, and textual specificity. To dynamically balance their influence, we introduce a dynamic modality weighting mechanism. Global representations of each modality are computed as

$$g_v = \text{GAP}(\boldsymbol{X}_i), \quad g_l = \text{GAP}(\boldsymbol{L}) \tag{12}$$

where GAP$(\cdot)$ denotes the global average pooling operation and g_v and g_l represent the global visual and linguistic embeddings used for modality importance assessment, respectively.

These representations are concatenated and fed into a lightweight multilayer perceptron to produce modality weights

$$[w_v, w_l] = \text{Softmax}\left(\boldsymbol{W}_2\sigma\left(\boldsymbol{W}_1\left(\left[g_v; g_l\right]\right)\right)\right) \tag{13}$$

where $[\cdot;\cdot]$ denotes concatenation along the channel dimension. $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are learnable projection matrices for modeling cross-modal interactions and estimating modality importance, and $\sigma(\cdot)$ denotes a nonlinear activation function. The modality weights satisfy $w_v + w_l = 1$, allowing the network to focus on the more informative modality for each sample dynamically. This design reduces the interference from less informative modalities and improves robustness in challenging scenarios.

3) Weighted Feature Fusion: The predicted modality weights are then applied to the bidirectional attention outputs

$$\tilde{\boldsymbol{A}}^i_{v\to l} = w_l \cdot \boldsymbol{A}^i_{v\to l}, \quad \tilde{\boldsymbol{A}}^i_{l\to v} = w_v \cdot \boldsymbol{A}^i_{l\to v}. \tag{14}$$

The weighted features are concatenated and passed through a spatial gating network to emphasize spatially informative regions and suppress irrelevant backgrounds

$$\boldsymbol{F}_s = \text{Spatial_Gate}\left(Conv\left[\tilde{\boldsymbol{A}}^i_{v\to l}; \tilde{\boldsymbol{A}}^i_{l\to v}\right]\right) \tag{15}$$

where Spatial_Gate$(\cdot)$ denotes a lightweight gating function composed of two linear projections with a ReLU activation in between, followed by a sigmoid function to generate spatially adaptive gating coefficients. The spatial gate (SG) adaptively modulates the fused features by highlighting spatial regions that exhibit strong visual–textual correspondence, thereby mitigating background interference in complex remote sensing scenes.

Finally, the output features of DMBM are obtained as

$$\boldsymbol{F}_{\text{DMBM}} = \boldsymbol{F}_s \otimes \boldsymbol{X}_i + \boldsymbol{X}_i \tag{16}$$

where the spatially gated fusion feature $\boldsymbol{F}_s$ modulates the original visual representation $\boldsymbol{X}_i$ in a residual manner. This formulation enables effective integration of visual cues and textual guidance while preserving the original spatial structure, which is particularly suited for referring segmentation in complex remote sensing scenes.

D. Loss Function and Implementation Details

The overall training objective integrates pixel-level supervision and patch-level representation learning to jointly optimize segmentation accuracy and cross-modal discriminability. Specifically, pixel-wise cross-entropy loss and Dice loss are adopted to supervise dense mask prediction, ensuring both semantic correctness and spatial completeness, while the S^2CL loss introduces region-aware contrastive constraints to enhance feature discrimination by suppressing background

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Fig. 3. Qualitative segmentation examples produced by the proposed method on the RefSegRS test dataset.

interference and strengthening object-level semantic consistency. The final loss function is defined as

$$\mathcal{L} = (1 - \lambda_{\text{Dice}})\,\mathcal{L}_{\text{CE}} + \lambda_{\text{Dice}}\mathcal{L}_{\text{Dice}} + \lambda_{\text{S}^2\text{CL}}\mathcal{L}_{\text{S}^2\text{CL}} \tag{17}$$

where λ_{Dice} and $\lambda_{\text{S}^2\text{CL}}$ denote the weighting factors for the Dice loss and S^{2}CL loss, respectively. In our experiments, λ_{Dice} is fixed to 0.1 following prior works [1], [2], [3], while $\lambda_{\text{S}^2\text{CL}}$ is empirically determined through ablation studies, as detailed in Section IV.

The proposed framework is implemented using the PyTorch library [51]. To evaluate the effectiveness and generality of the proposed method, we consider two image–text encoding settings. The first adopts a CLIP-based encoder, where visual and textual branches are jointly pretrained on large-scale image–text pairs, providing strong cross-modal priors. The second employs decoupled encoders, using a Swin Transformer pretrained on ImageNet-22K [52] for image encoding and a BERT model [53] for text encoding, representing a non-CLIP setting. In both cases, the encoders are fine-tuned end-to-end on the target datasets. Experiments under these two settings demonstrate that the proposed method consistently improves performance across different backbone architectures and pretraining paradigms, validating its robustness and encoder-agnostic design.

Experiments are conducted on three widely used RRSIS benchmarks, namely RefSegRS [1], RRSIS-D [2], and RISBench [12]. All input images are uniformly resized to a spatial resolution of 480×480 pixels for efficient training and evaluation. The model is trained for 60 epochs with an initial learning rate of 5×10^{-5}. During optimization, the AdamW optimizer [54] is employed, with a weight decay coefficient of 0.1 to regularize model parameters and mitigate overfitting. All experiments are carried out on an NVIDIA GeForce RTX 4090 GPU, and the batch size is fixed to 8 across all settings.

IV. Experiments

A. Experimental Dataset and Metrics

To validate the effectiveness of the proposed method, experiments are conducted on three publicly available benchmarks, namely RefSegRS [1], RRSIS-D [2], and RISBench [12]. These datasets are widely adopted in the RRSIS community, providing standardized evaluation protocols and diverse scene coverage.

1) RefSegRS [1] consists of 4420 image–text–mask triplets, which are split into 2172 samples for training, 431 for validation, and 1817 for test. The dataset covers 14 object categories and each referring expression is characterized by five attributes to describe object appearance. All images have a fixed spatial size of 512×512 pixels, with a spatial resolution of 0.13 m/pixel.
2) RRSIS-D [2] contains 17 402 annotated samples, among which 12 181 are used for training, 1740 for validation, and 3481 for test. This dataset includes 20 object categories and exhibits greater diversity in both scene scale and spatial resolution compared with RefSegRS, with

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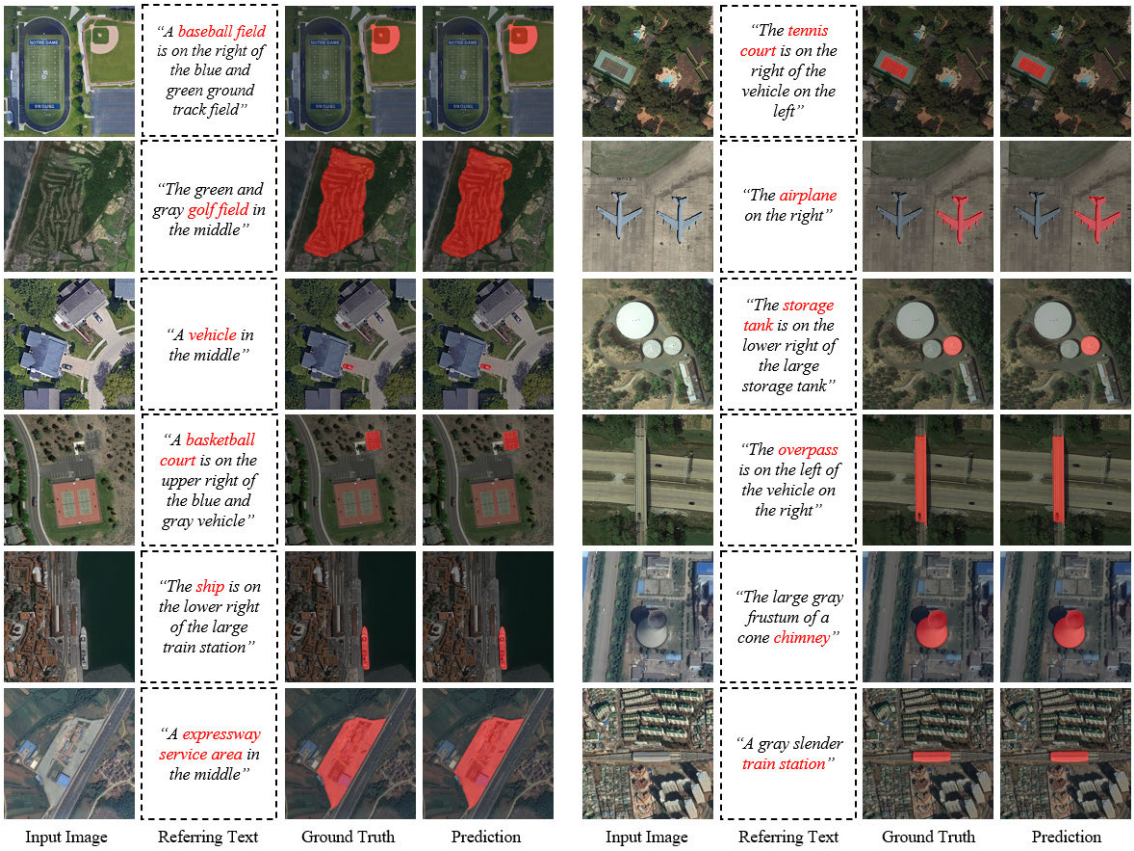


Fig. 4. Qualitative segmentation examples produced by the proposed method on the RRSIS-D test dataset.

TABLE I

RESULTS OF REFERRING IMAGE SEGMENTATION WITH DIFFERENT METHODS ON THE **REFSEGRS DATASET**. THE BEST PERFORMANCE IS BOLD

Methods	Publication	Image Enc.	Text Enc.	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	oIoU
LSTM-CNN [23]	ECCV2016	VGG-16	LSTM	15.69	10.57	5.17	1.10	0.28	24.76	53.83
ConvLSTM [26]	CVPR2018	ResNet-101	LSTM	31.21	23.39	15.30	7.59	1.10	43.34	66.12
BRINet [32]	CVPR2020	ResNet-101	LSTM	22.56	15.74	9.85	3.52	0.50	32.87	60.16
LAVT [55]	CVPR2022	Swin-B	BERT	70.23	55.53	30.05	14.42	4.07	57.30	76.21
CrossVLT [56]	TMM2023	Swin-B	BERT	71.16	58.28	34.51	16.35	5.06	58.84	77.44
RMSIN [2]	CVPR2024	Swin-B	BERT	71.60	55.97	31.87	11.72	1.93	57.78	71.73
LGCE [1]	TGRS2024	Swin-B	BERT	76.55	67.03	44.85	19.04	5.67	61.90	77.62
FIANet [3]	TGRS2025	Swin-B	BERT	84.09	77.05	61.86	33.41	7.10	68.67	78.32
Enti-TwistNet [16]	TGRS2025	Swin-B	BERT	77.72	69.30	45.01	26.66	8.49	63.82	78.98
STDNet [43]	JSTARS2025	Swin-B	BERT	83.60	75.07	62.69	34.40	9.14	67.95	80.57
CroBIM-U [44]	arXiv2026	ConvNeXt-B	BERT	76.68	62.54	34.81	13.50	3.26	60.08	73.81
Proposed	-	Swin-B	BERT	86.35	80.17	70.01	45.95	11.78	72.10	80.97
Proposed	-	CLIP	CLIP	**86.46**	**81.07**	**72.92**	**54.65**	**18.44**	**73.75**	**81.54**

image sizes of 800 × 800 pixels and spatial resolutions ranging from 0.5 to 30 m/pixel.

3) RISBench [12] is a large-scale and challenging dataset, containing 52 472 annotated image–text–mask triplets with 26 300/10 013/16 158 samples for training, validation, and test, respectively. It covers 26 object categories and associates each referring text with eight attributes. The dataset provides images with a fixed size of 800 × 800 pixels and the range of spatial resolutions is from 0.1 to 30 m/pixel.

For quantitative evaluation, we follow some prior works [1], [2], [55] and adopt overall intersection over union (oIoU) and mean intersection over union (mIoU) as the primary performance metrics. The oIoU aggregates intersection and union areas over the entire test set, whereas mIoU computes the average IoU across individual samples, treating objects of different scales equally. In addition, precision at multiple IoU thresholds (Pr@X, where $X \in 0.5, 0.6, 0.7, 0.8, 0.9$) is reported to measure the proportion of test samples whose predicted masks exceed a given IoU threshold.

B. Comparisons With Other Methods

We evaluate the proposed method compared with some state-of-the-art methods on the RefSegRS, RRSIS-D, and

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TABLE II
RESULTS ON EACH CATEGORY OF **REFSEGRS DATASET**. THE BEST PERFORMANCE IS BOLD

category	RMSIN	LGCE	FIANet	Ours
road	66.67	74.03	74.13	**76.42**
vehicle	58.66	63.05	70.15	**75.25**
car	57.63	60.79	68.55	**74.38**
van	47.09	41.60	61.06	**70.19**
building	76.84	81.9	81.34	**84.36**
truck	51.92	62.69	74.48	**80.42**
trailer	61.65	49.82	**74.92**	74.77
bus	60.20	45.36	**72.40**	71.11
road marking	18.60	6.66	22.85	**24.05**
bikeway	50.35	54.23	61.16	**67.41**
sidewalk	49.12	61.68	62.90	**66.28**
tree	49.82	67.68	**83.75**	81.69
low vegetation	43.73	43.68	**44.84**	39.06
impervious surface	76.55	83.18	81.53	**84.76**
average	54.92	56.89	66.72	**69.30**

RISBench datasets. The compared methods include the ones specifically developed for remote sensing scenarios, such as LGCE [1], RMSIN [2], FIANet [3], CDFormer [14], and Enti-TwistNet [16], and the others are designed for natural image referring segmentation. Quantitative results on these benchmarks are summarized in Tables I–V, and qualitative visualizations on these three datasets are shown in Figs. 3–5.

1) Quantitative Results on the RefSegRS Dataset: Table I reports the quantitative comparison of different methods on the RefSegRS test dataset. The proposed approach, with the setting of CLIP as image and text encoders, consistently achieves the best performance across all evaluation metrics, highlighting the effectiveness of the proposed design. To further analyze performance at a finer granularity, we present class-wise mIoU comparisons across the 14 scene categories in RefSegRS, as summarized in Table II. As observed, segmentation difficulty varies substantially among different ground objects. Categories such as road marking remain challenging due to their thin structures and weak visual contrast, whereas impervious surface is much easier to segment. The proposed method achieves the best performance on most object categories. In particular, it delivers notable improvements over FIANet on categories such as vehicle, truck, and van, demonstrating the effectiveness of the proposed framework in distinguishing objects with diverse appearances and complex surrounding contexts. For the few categories where our method does not achieve the best performance, including low vegetation, tree, and bus, the performance gaps remain relatively small. These categories are inherently challenging due to ambiguous object boundaries, high visual similarity to surrounding regions, and considerable appearance variations, which make precise visual-language alignment more difficult. Despite these variations, the average mIoU of our method significantly exceeds that of all competing approaches, further validating its robustness and effectiveness in handling diverse object types within remote sensing scenes.

2) Quantitative Results on the RRSIS-D Dataset: The RRSIS-D dataset constitutes a more challenging benchmark than RefSegRS, featuring a larger scale, increased category diversity, and substantial variations in object size and spatial resolution. The overall quantitative comparison is summarized in Table III. As reported in Table III, under the configuration using Swin-B as the image encoder and BERT as the text encoder, the proposed method demonstrates superior performance on several evaluation metrics. Specifically, it achieves the best results in terms of oIoU and precision at higher IoU thresholds (Pr@0.6 to Pr@0.9). In addition to the overall evaluation, we further report class-wise segmentation performance on the 20 ground object categories of RRSIS-D in Table IV. Due to the pronounced scale diversity and heterogeneous spatial distributions of objects, this dataset poses greater challenges for accurate referring segmentation. Nevertheless, the proposed method delivers superior performance on the majority of categories, including road, vehicle, car, and van.

3) Quantitative Results on the RISBench Dataset: Compared with the RefSegRS and RRSIS-D datasets, RISBench expands the ground object categories to 26 classes and introduces eight semantic attributes while covering a much broader range of spatial resolutions, which makes it a more challenging benchmark. As reported in Table V, the proposed method, using Swin-B as the image encoder and BERT as the text encoder, achieves the best performance in terms of both oIoU and mIoU, as well as precision under different IoU thresholds (Pr@0.5 to Pr@0.9), outperforming recently proposed methods such as FIANet, Enti-TwistNet, and CDFormer. These results further demonstrate the effectiveness of the proposed approach in handling diverse ground object categories and complex referring expressions.

4) Qualitative Visualizations: We present representative qualitative results of the proposed method on the RefSegRS, RRSIS-D, and RISBench to provide an intuitive evaluation of its segmentation capability, as shown in Figs. 3–5. The visualizations include input images, the corresponding referring expressions, ground-truth masks, and predicted segmentation results. As illustrated, the proposed method demonstrates reliable localization and segmentation performance in complex remote sensing scenes across diverse object categories and scales, ranging from small objects (e.g., buses, trucks, and vehicles) to medium-sized objects (e.g., buildings, chimneys, and overpasses), and large-scale targets (e.g., expressway service areas and golf fields). In particular, the model demonstrates strong robustness in handling scenes with cluttered backgrounds and visually similar objects. Moreover, it successfully interprets spatial relational descriptions embedded in the referring expressions, such as "on the upper right" or "in the middle," and produces segmentation masks that are spatially consistent with the linguistic cues. These qualitative results further verify that the proposed method can effectively exploit multimodal information to distinguish target objects from surrounding background regions, highlighting its effectiveness and generalization capability.

From Tables I, III, and V, we observe that the CLIP-based backbone achieves better performance on RefSegRS, whereas the Swin-B/BERT backbone performs better on RRSIS-D and RISBench. We attribute this difference primarily to the distinct representation characteristics of the two vision–language backbones. Specifically, the CLIP-based backbone benefits

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TABLE III
RESULTS OF REFERRING IMAGE SEGMENTATION WITH DIFFERENT METHODS ON THE **RRSIS-D DATASET**. THE BEST PERFORMANCE IS BOLD

Methods	Publication	Image Enc.	Text Enc.	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	oIoU
RRN [26]	CVPR2018	ResNet-101	LSTM	51.07	42.11	32.77	21.57	6.37	45.64	66.43
LSCM [57]	ECCV2020	ResNet-101	LSTM	56.02	46.25	37.70	25.28	8.27	49.92	69.05
CMPC [58]	CVPR2020	ResNet-101	LSTM	55.83	47.40	36.94	25.45	9.19	49.24	69.22
BRINet [32]	CVPR2020	ResNet-101	LSTM	56.90	48.77	39.12	27.03	8.73	49.65	69.88
CMPC+ [59]	TPAMI2021	ResNet-101	LSTM	57.65	47.51	36.97	24.33	7.78	50.24	68.64
LAVT [55]	CVPR2022	Swin-B	BERT	66.93	60.99	51.71	39.79	23.99	59.05	76.58
CrossVLT [56]	TMM2023	Swin-B	BERT	70.38	63.83	52.86	42.11	25.02	61.00	76.32
LGCE [1]	TGRS2024	Swin-B	BERT	69.41	63.06	53.46	41.22	24.27	61.02	76.24
RMSIN [2]	CVPR2024	Swin-B	BERT	71.96	65.76	55.16	42.03	25.02	62.27	76.50
FIANet [3]	TGRS2025	Swin-B	BERT	74.46	66.96	56.31	42.83	24.13	64.01	76.91
CDFormer [14]	JSTARS2025	Swin-B	BERT	74.20	67.62	55.59	42.37	23.59	63.77	77.26
CroBIM-U [44]	arXiv2026	ConvNeXt-B	BERT	**75.60**	67.68	56.47	42.57	24.16	**65.07**	76.70
Proposed	-	Swin-B	BERT	74.89	**68.89**	**56.68**	**44.13**	**25.60**	64.49	**77.69**
Proposed	-	CLIP	CLIP	72.54	66.39	55.79	41.17	23.38	62.82	75.26

TABLE IV
RESULTS ON EACH CATEGORY OF **RRSIS-D DATASET**. THE BEST PERFORMANCE IS BOLD

category	RMSIN	LGCE	FIANet	Ours
airplane	68.08	68.11	68.66	**69.98**
airport	56.11	56.43	57.07	**59.81**
golf field	76.68	77.19	**77.35**	76.62
expressway service area	66.93	70.93	70.44	**71.46**
baseball field	83.09	84.90	84.87	**85.44**
stadium	81.91	82.54	82.00	**84.06**
ground track field	73.65	73.33	**76.99**	76.39
storage tank	72.26	74.37	**74.86**	74.74
basketball court	68.42	68.44	**68.41**	68.11
chimney	76.68	75.63	78.48	**79.91**
tennis court	70.14	67.67	70.01	**71.09**
overpass	62.67	58.19	61.30	**62.74**
train station	64.64	63.48	**65.96**	65.32
ship	65.71	61.63	64.82	**67.19**
expressway toll station	68.70	64.54	**71.31**	69.48
dam	60.40	60.47	62.03	**62.29**
harbor	36.74	34.24	37.94	**38.12**
bridge	47.63	43.12	**49.66**	49.26
vehicle	41.99	40.76	46.72	**46.80**
windmill	60.18	56.51	60.25	**60.80**
average	65.13	64.12	66.46	**66.98**

from large-scale vision–language pretraining, providing strong semantic alignment and superior transferability on datasets with limited training samples, such as RefSegRS. In contrast, the Swin-B/BERT framework employs independent visual and textual encoders, enabling more detailed spatial modeling and language understanding when sufficient training data are available, as in RRSIS-D and RISBench.

C. Ablation Studies

We perform a series of ablation experiments on the RefSegRS test set to evaluate the contributions of the key components of the proposed method.

1) Effectiveness of S^2CL and DMBM: We conduct comprehensive ablation experiments under different image–text encoder configurations to evaluate the effectiveness of S^2CL and DMBM. The results on the RefSegRS test set are summarized in Table VI. For the CLIP-based encoder setting, Model-D serves as the baseline. Introducing the S^2CL (Model-E) leads to a clear performance improvement across almost all evaluation metrics, demonstrating the effectiveness of structure-constrained semantic contrastive supervision. Further incorporating the DMBM yields the best overall results, indicating that these two are complementary. To further assess the generality of the proposed components, we repeat the same ablation study using Swin-B and BERT as the image and text encoders, respectively. As shown in Table VI, consistent performance gains are also observed on this non-CLIP backbone, even though it already provides a strong baseline. These results confirm that both S^2CL and DMBM are encoder-agnostic and can be effectively applied across different backbone architectures.

In addition, we perform comparative experiments to analyze the performance difference between conventional image-level CL and the proposed S^2CL. The quantitative results are reported in Fig. 6. As shown in this figure, S^2CL consistently outperforms standard image-level CL across all evaluation metrics, demonstrating its superiority in learning finer-grained and semantically aligned representations for RRSIS.

2) Different Designs of S^2CL: To analyze the contribution of each component in the S^2CL, we conduct an ablation study by individually removing key modules from the complete S^2CL framework. The quantitative results are reported in Table VII. Removing mask guidance, visual feature selection (VFS), or HNSW consistently degrades performance, reducing mIoU to 71.69%, 71.66%, and 71.63%, respectively. These drops verify the effectiveness of the corresponding components in suppressing background noise, eliminating redundant visual information, and refining feature discrimination. Notably, removing the GP leads to the most pronounced decline in both mIoU and oIoU, highlighting its critical role in enabling effective patch-level CL. These results demonstrate that all components collaboratively contribute to the overall performance of S^2CL.

3) Different Designs of DMBM: To investigate the contribution of each component in the DMBM, we perform a detailed ablation study, with the results summarized in Table VIII. Compared to the baseline with unidirectional cross-modal

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TABLE V

RESULTS OF REFERRING IMAGE SEGMENTATION WITH DIFFERENT METHODS ON THE **RISBENCH DATASET**. THE BEST PERFORMANCE IS BOLD

Methods	Publication	Image Enc.	Text Enc.	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	oIoU
RRN [26]	CVPR2018	ResNet-101	LSTM	55.04	47.31	39.86	32.58	13.24	43.18	49.67
LSCM [57]	ECCV2020	ResNet-101	LSTM	55.26	47.14	40.10	33.29	13.91	43.69	50.08
CMPC [58]	CVPR2020	ResNet-101	LSTM	55.17	47.84	40.28	32.87	14.55	43.82	50.24
BRINet [32]	CVPR2020	ResNet-101	LSTM	52.87	45.39	38.64	30.79	11.86	42.91	48.73
CMPC+ [59]	TPAMI2021	ResNet-101	LSTM	58.02	49.00	42.53	35.26	17.88	46.73	53.98
CRIS [22]	CVPR2022	CLIP	CLIP	63.67	55.73	44.42	28.80	13.27	55.18	69.11
LAVT [55]	CVPR2022	Swin-B	BERT	69.40	63.66	56.10	44.95	25.21	61.93	74.15
CrossVLT [56]	TMM2023	Swin-B	BERT	70.62	65.05	57.40	45.80	26.10	62.84	74.33
CARIS [60]	ACM MM2023	Swin-B	BERT	73.94	68.93	62.08	50.31	29.08	65.79	75.10
LGCE [1]	TGRS2024	Swin-B	BERT	69.64	64.07	56.26	44.92	25.74	62.13	73.87
RMSIN [2]	CVPR2024	Swin-B	BERT	71.01	65.46	57.69	45.50	25.92	63.07	74.09
FIANet [3]	TGRS2025	Swin-B	BERT	75.10	70.27	63.82	53.15	32.24	67.17	73.75
Enti-TwistNet [16]	TGRS2025	Swin-B	BERT	76.13	71.47	63.48	52.38	27.46	67.50	74.07
CDFormer [14]	JSTARS2025	Swin-B	BERT	76.27	71.35	64.21	52.91	31.59	67.80	74.34
Proposed	-	Swin-B	BERT	**76.79**	**71.91**	**65.18**	**54.77**	**34.51**	**68.54**	**75.28**
Proposed	-	CLIP	CLIP	74.81	69.80	62.90	51.47	31.06	66.84	72.00

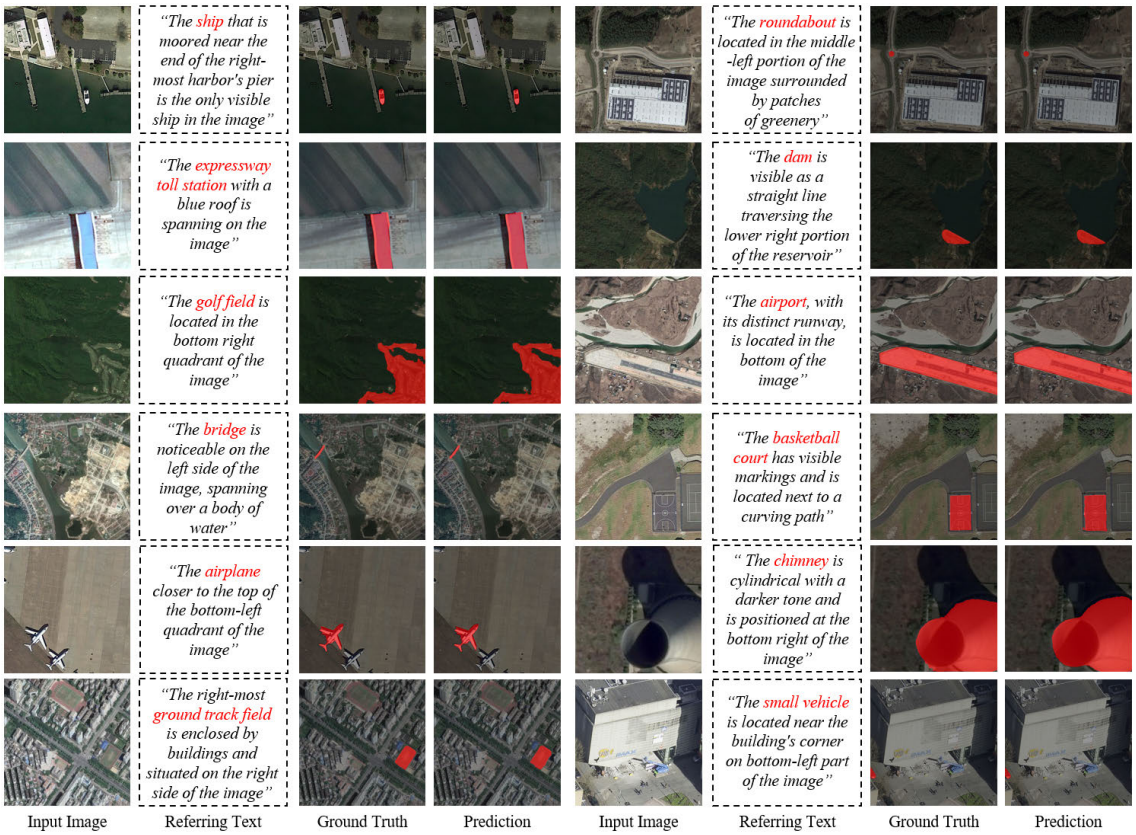


Fig. 5. Qualitative segmentation examples produced by the proposed method on the RISBench test dataset.

TABLE VI

ABLATION STUDIES ABOUT THE S^{2}CL AND DMBM WITH DIFFERENT ENCODERS ON THE REFSEGRS TEST SET

Method	Image Enc.	Text Enc.	S^{2}CL	DMBM	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	oIoU
Model-A	Swin-B	BERT	×	×	85.97	79.64	68.57	43.15	10.79	71.32	81.18
Model-B	Swin-B	BERT	✓	×	85.97	79.80	69.01	43.48	10.35	71.33	81.32
Model-C (ours)	Swin-B	BERT	✓	✓	86.35	80.17	70.01	45.95	11.78	72.10	80.97
Model-D	CLIP	CLIP	×	×	86.02	78.70	66.54	37.75	7.98	70.19	79.14
Model-E	CLIP	CLIP	✓	×	86.58	80.08	69.95	48.98	11.56	72.02	80.86
Model-F (ours)	CLIP	CLIP	✓	✓	86.46	81.07	72.92	54.65	18.44	73.75	81.54

attention (UCA), introducing BCA leads to a clear performance gain, improving oIoU to 80.35%, which highlights the importance of bidirectional visual-language interaction. Incorporating the SG further yields modest improvements by

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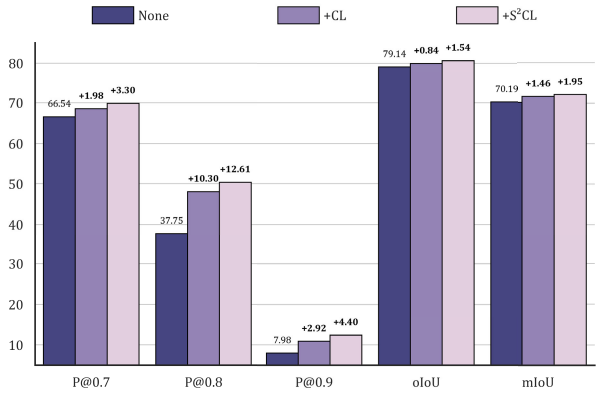


Fig. 6. Comparisons of the conventional CL and the proposed S^{2}CL.

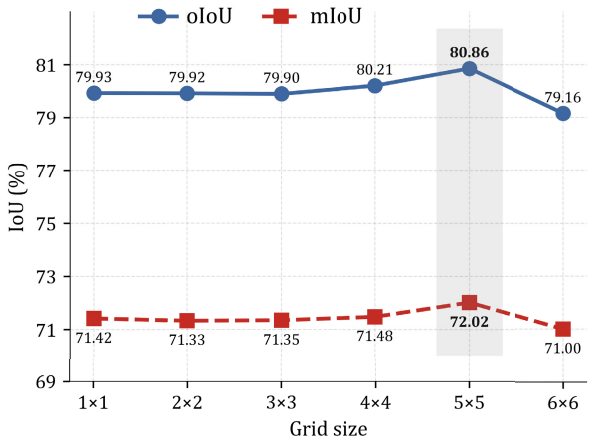


Fig. 7. Quantitative comparisons of different grid sizes used in S^{2}CL.

TABLE VII

ABLATION STUDIES ON THE DIFFERENT DESIGNS OF S^{2}CL

Method	Pr@0.5	Pr@0.7	Pr@0.9	mIoU	oIoU
w.o Mask	85.31	69.62	10.73	71.69	80.36
w.o GP	85.03	68.13	11.45	71.42	79.93
w.o VFS	85.91	69.07	11.78	71.66	80.34
w.o HNSW	85.31	68.79	12.00	71.63	80.26
Full (ours)	86.58	69.95	11.56	72.02	80.86

TABLE VIII

ABLATION STUDIES ON THE DIFFERENT DESIGNS OF DMBM

Method	Pr@0.5	Pr@0.7	Pr@0.9	mIoU	oIoU
UCA (baseline)	86.02	66.54	7.98	70.19	79.14
BCA	86.30	70.34	16.95	73.21	80.35
BCA + SG	86.20	70.45	17.02	73.16	80.42
BCA + SG + DMWG	86.74	71.71	18.21	73.29	80.98

TABLE IX

EFFECT OF DIFFERENT WEIGHT COEFFICIENTS OF S^{2}CL LOSS

λ_{S^2CL}	Pr@0.5	Pr@0.7	Pr@0.9	mIoU	oIoU
0.0	86.02	66.54	7.98	70.19	79.14
0.1	85.14	67.91	10.79	71.28	79.72
0.2	86.58	69.95	11.56	72.02	80.86
0.3	85.47	69.84	11.50	71.93	80.75
0.6	85.69	68.08	12.22	71.53	80.07
1.0	85.75	66.26	8.15	70.37	78.84

enabling spatially adaptive feature modulation. Finally, integrating dynamic modality weight generation (DMWG) brings the most notable additional gain, as it dynamically balances visual and textual contributions across samples, resulting in the highest oIoU of 80.98%.

4) Influence of S^{2}CL Weight: To examine the effect of the weighting factor of the S^{2}CL loss in the overall training objective, we conduct ablation experiments with different values of λ_{S^2CL}, where $\lambda_{S^2CL} = 0$ denotes the baseline without the S^{2}CL loss, trained with pixel-level supervision only. As shown in Table IX, the performance varies with different weight settings, reflecting the influence of the S^{2}CL term on optimization. The best result is obtained when λ_{S^2CL} is set to 0.2, achieving the oIoU of 80.86% and mIoU of 72.02%. This observation indicates that assigning an appropriate weight to the S^{2}CL loss effectively enhances patch-level discriminability without overwhelming pixel-level supervision.

5) Influence of GP Size: To analyze the effect of GP size in S^{2}CL, we conduct ablation experiments with different grid sizes and report oIoU and mIoU in Fig. 7. With a coarse 1×1 partition, the model achieves an oIoU of 79.93%, indicating limited benefit from patch-level contrastive supervision. Increasing the grid size gradually improves performance, which peaks at 5×5 with an oIoU of 80.68%. This setting provides a good balance between preserving regional semantics and enabling fine-grained CL. However, further increasing the grid size to 6×6 leads to performance degradation, due to excessive fragmentation of semantic regions.

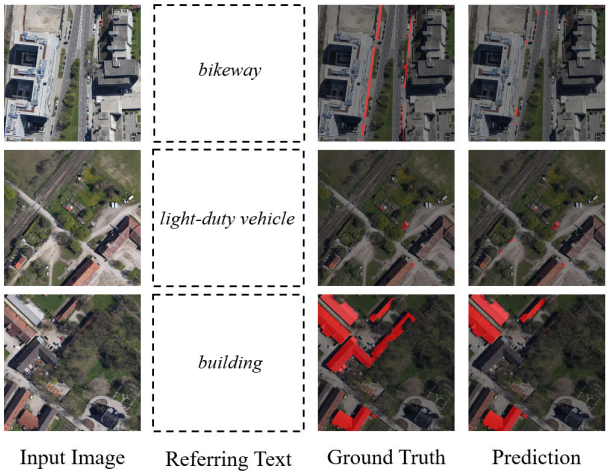


Fig. 8. Some failure cases of the proposed method in the RefSegRS test data.

D. Discussion of Failure Cases

It presents some failure cases of the proposed method on the RefSegRS test set, as shown in Fig. 8. The failure cases mainly occur in challenging scenarios involving low visual contrast with complex backgrounds. In Fig. 8, the ground objects are relatively difficult to segment when the target regions have weak appearance differences from surrounding areas or lack sufficient discriminative visual pat-

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terns. Moreover, objects with diverse spatial scales and unclear boundaries may hinder the model from accurately capturing fine-grained correspondence between visual features and referring expressions. These failure cases indicate that, although the proposed method effectively improves cross-modal representation learning, accurately interpreting subtle visual cues and ambiguous spatial structures in complex remote sensing scenes remains challenging. Future studies may further explore more effective mechanisms for enhancing fine-grained spatial-semantic reasoning and improving the robustness of cross-modal representation learning under low-contrast and scale-diverse scenarios.

V. Conclusion

In this article, we propose a novel approach for RRSIS. We introduce an S^{2}CL framework, named S^{2}CLNet, which leverages referring masks to guide patch-level representation learning, promoting fine-grained alignment between visual regions and textual descriptions. In addition, a DMBM is designed to dynamically integrate visual and textual features, accounting for their complementary contributions and sample-specific importance. By combining S^{2}CL and DMBM, the proposed method enhances local semantic discrimination, spatial consistency, and cross-modal feature interaction, which is especially beneficial for complex remote sensing scenes with diverse object categories, large-scale variations, and spatial relational descriptions. Extensive experiments on three public RRSIS benchmarks, including RefSegRS, RRSIS-D, and RISBench, demonstrate that our approach achieves superior performance compared with existing methods, and ablation studies validate the effectiveness of both S^{2}CL and DMBM.

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